



Discovering and evaluating organizational knowledge from textual data: Application to crisis management[☆]

Dhouha Grissa^{*}, Eric Andonoff, Chihab Hanachi

Department of Computer Science, Toulouse 1 Capitole University, Institut de Recherche en Informatique de Toulouse (IRIT) - CNRS Laboratory, 2 rue du Doyen Gabriel Marty, 31042 Toulouse Cedex, France

ARTICLE INFO

Keywords:

Crisis management
Knowledge discovery
Multi-agent system
Text mining
Association rules
Organizational structure
Evaluation

ABSTRACT

Crisis management effectiveness relies mainly on the quality of the distributed human organization deployed for saving lives, limiting damage and reducing risks. Organizations set up in this context are not always predefined and static; they could evolve and new forms could emerge since actors, such as volunteers or NGO, could join dynamically to collaborate. To improve crisis resolution effectiveness, it is first important to understand, analyze and evaluate such dynamic organizations in order to adjust crisis management plans and ease coordination among actors. Giving a textual experience feedback from past crisis, the objective of this paper is to discover the organizational structure deployed in the considered crisis and then evaluate it according to a set of criteria. For that purpose, we combine in a coherent framework text and association rule mining for pattern discovery and annotation, and multi-agent system models and techniques for formally building and evaluating organizational structures. We present the *OSminer* algorithm that discovers association rules based on relevant textual patterns and then builds an organizational structure including three main relations between actors: power, control and coordination. A real-life case study, a flood crisis hitting the south west of France, serves as a basis for testing/experimenting our solution. The organizational structure, discovered in this case study, has 24 actors. Its evaluation indicates its efficiency, but shows that it is neither robust nor flexible. Our findings highlight the potential of our approach to discover and evaluate organizational structures from a text recording interactions between stakeholders in a crisis context.

1. Introduction

The effectiveness of crisis management (flood, earthquake, tsunami...) is strongly related to the quality of the human organization deployed for saving lives, limiting damage and reducing risks. The importance of an organization in this context results from the fact that, it fixes the roles that could be played by the actors in the field, determines their relationships (power, delegation, coordination, control, ...), regulates their interactions [1] and eases the coordination of their actions [2]. Sometimes, the organization is pre-determined by crisis resolution plans, other times it emerges from interactions between actors, especially when unplanned actors (citizens, NGOs¹) volunteer. Even when the organization is pre-determined the real-life/concrete organization set up in the field

[☆] This work was supported by the French National Research Agency - Project "Génépi" about granularity of control levels in Crisis Management under grant agreement ANR-14-CE28-0029.

^{*} Correspondence to: Center for non-coding RNA in Technology and Health, Department of Veterinary and Animal Science, University of Copenhagen, 1870, Frederiksberg C, Denmark.

E-mail addresses: dhouha.grissa@sund.ku.dk (D. Grissa), Eric.Andonoff@ut-capitole.fr (E. Andonoff), Chihab.Hanachi@ut-capitole.fr (C. Hanachi).

¹ National Governmental Organization.

may evolve and be different from the one planned. This could be notably due to unforeseeable events or the variable availability and efficiency of material and human resources involved in each crisis resolution. To improve organizational effectiveness, in such highly dynamic environments, there is first, an urgent need to understand how past crisis have been managed from an organizational point of view.

In this paper, we are interested in the discovery and evaluation of concrete organizations, i.e. organizations effectively deployed in crisis management, by analyzing experience feedback recorded in textual documents. For that purpose, we propose a three-steps process approach. It first exploits text and association rules mining for information extraction and annotation. Then, the "OSminer" algorithm (OS stands for Organizational Structure) discovers the real organizational structure. Here, we define an organizational structure as a formal and computational representation of an organization, in terms of directed graphs. Finally, we evaluate the discovered organizational structure in terms of robustness, efficiency and flexibility.

More precisely, the objective of this paper is to present a tooled and reusable approach able to explore a crisis textual experience feedback in order to discover and evaluate organizations set up in this context. The proposed approach is directly related to research work performed in the fields of Multi-Agent Systems (MAS), and text and data mining. Indeed, the evaluation of Organizational Structures (OS) is based on the research work performed by Grossi [1,3]. In this work, Grossi abstracts agent organizations by graphs representing relations (edges) of power, control and coordination between roles (nodes) of the considered organization and assesses their performance using properties of robustness, flexibility and efficiency [3]. The discovery of agent organizations in crisis management has already been studied in [2] but from a log file of activities where the actors and their activities are highlighted. It assumes the existence of a specific tool recording actions performed in the crisis field by actors. In the majority of cases, such a tool does not exist. However, experience feedback is often recorded in textual documents at the end of a crisis with the aim to learn from this past experience. But such documents are rarely automatically treated for discovering models related to organizations or plans. In this paper, we try to fill this gap by combining text mining, association rule mining and multi-agent systems fields to provide means to discover, understand and evaluate organizations from textual records. It is a step forward with respect to the state of the art in crisis and disaster management.

This research work includes four main contributions. The first one is the process that transforms text records into an organizational structure (OS) patterns. Based on pattern mining, we explore experience feedback recorded in textual documents [4] to extract relevant textual patterns useful to mine associations between actors regarding the three relations of power, control and coordination as defined in [1]. The second contribution consists in proposing a new interactive approach [5,6] to discover the corresponding organizational structure, based on mining of association rules and Osminer algorithm to uncover how actors interact in a real crisis management case. This is a promising idea since mining association rules enable not only identifying the actors who interact and how they do it, but it also highlights the central actors of the organization. The third contribution extends the work of Grossi [3] for evaluating the performance of an OS, by introducing additional metrics based on graph theory. These metrics exploit different levels of the organization, as role, actor, etc. The last contribution is an evaluation of our approach that consists in assessing a concrete organization derived from a real-world crisis management example. This evaluation could be used as a basis to provide organizational recommendations to the authorities in order to better address future crisis.

The novelty aspect of this paper is to combine two research areas (text and data mining with multi-agent systems) for discovering, representing, understanding and evaluating human organizations from textual experience feedback. Also, the proposed approach is applied to a real world crisis, i.e. a flood management in the southwest of France for which we have an available text describing its management. Nevertheless, our approach is general enough to be exploited in other types of crisis and more generally in any distributed application domains where organizations are emergent and should be understood and therefore discovered, represented and evaluated.

This paper is organized as follows. Section 2 introduces the theoretical background and presents related works. Section 3 presents our approach through its three steps: the extraction of the significant patterns, the discovery of the organizational structure and its evaluation, thus introducing the three first contributions. Section 4 describes the last contribution by presenting a real case study and its treatment following our approach. We visualize, validate, and discuss the results. Finally, Section 5 concludes the paper and gives some directions for future works.

2. Related work

Herein, we position our contribution w.r.t. to the state of the art. We focus on two main complementary points of view: (i) text and association rule mining for crisis management, and (ii) organizational structure discovery and evaluation.

Regarding text and data mining, they are both important research fields included in the domain of information mining [7]. Many researchers focused on adapting data mining techniques to extract useful information from documents, as association rules [8–11].

We chose association rules technique as it allows identifying co-occurrences and interesting connections between different items in a database. In our case, it would help us to discover central actors as well as the different interactions between them. The original combination use of association rules and text mining allows us to first identify the different textual patterns (text mining) and then discover their relations (association rules).

An association rule [12,13] in the context of text mining is based on a set I of keywords or items extracted from a collection of indexed paragraphs $P = \{P_1, P_2, \dots, P_m\}$ in a given document D . It is defined by:

$$X \Rightarrow Y \quad (1)$$

where, each paragraph is a set of keywords such that $P_i \subseteq I$, X and Y are sets of disjoint keywords (called itemsets or patterns) from I , i.e., $X, Y \subseteq I$ and $X \cap Y = \emptyset$. A paragraph P_i is said to contain X_i if and only if $X_i \subseteq P_i$. X represents the antecedent of the rule, and Y is the consequent. An association rule is an implication expression of the form "if condition, then result". The predominantly used evaluation method of such rule is essentially based on the notions of both quality (interestingness) measures, confidence and support. The support of an association rule $X \Rightarrow Y$ in I indicates its frequency of occurrence $Sup(X \cup Y)$, i.e., it indicates how frequently the items in the rule occur together. The confidence is the conditional probability $P(Y/X)$ that the consequent Y will occur given the occurrence of the antecedent X . They are respectively denoted as:

$$Sup(X \Rightarrow Y) = \frac{\text{Support count of } XY}{\text{Total number of paragraphs } P}; \quad (2)$$

$$Conf(X \Rightarrow Y) = \frac{Support(XY)}{Support(X)} \quad (3)$$

Mining Association rules consist in extracting strong rules, with support and confidence values greater than given thresholds: minimum support (minsup) and minimum confidence (minconf). Such rules represent regularities, correlations, and associations discovered from indexed paragraphs of a given document [14].

For example, given a set of words frequently used together in the field of multi-agent systems, {agent, cooperation, department}, we could derive the following rule *agent, department* $\Rightarrow$ *cooperation* with a support and confidence values higher than fixed thresholds.

Dealing with structured data, the use of data mining techniques [15] needs to be adjusted to analyze unstructured textual data. In this context, authors in [11] studied huge number of text documents, in the form of WebPages containing information news about the outbreak of the bird flu disease. They used association rules in text mining to highlight correlations between features in the texts, and introduced a system for automatically extracting association rules from WebPages news documents. In [10], authors used association rules for personalized knowledge discovery from text. They evaluated the semantic distance between the antecedent and the consequent of a rule in the background knowledge. A recent study [16] encompassed the use of association rule mining technique on both types of data i.e., free-text and categorical data to extract interesting inherent relationships hidden in accident data sets obtained from an integrated steel plant in India. The textual data consists in a narrative field describing incidents such as causes and contributory factors that authors were looking to uncover via the use of association rules. In [17,18], a MAS serves as a support for association rule extraction. In [17], authors recommended a supervised extraction with an expert domain and advocated a cooperative approach for this extraction. In [18], a multi-criteria analysis approach is recommended for such extraction. Association rules were also applied in bioinformatics, as in [19] to discover COVID-19 symptom patterns and rules that would support the initial identification of severe COVID-19 cases, or in [20] to identify symptoms and risk factors among three main prevalent diseases (cardiovascular disease, hepatitis and breast cancer). Also, they are used as a solution to avoid financial crisis by discovering forecasting changes in Korea composite Stock Price Index [21]. In addition to these studies and to support the analyses of a large number of association rules with multiple antecedents extracted from a large text corpus, researchers presented novel visualization techniques that were applied on a text mining system [22,23]. For example in [24], authors presented a three-dimensional (3D) display of association rules with minimum human interaction. Text mining techniques were also used to analyze experience feedback as performed by [25], who applied statistical and visualization techniques (the frequency of the words and word mapping for associations finding and visualization) to consolidate staff experience feedback survey conducted in a hospital in New York. In the context of crisis management, text mining has been mainly used for analyzing social media and particularly interactions involving citizens, volunteers and/or institutions for categorizing topics of interest or deduce phenomena (rumor, panics, etc.) or social groups (e.g., community) [26].

This brief overview highlights that text mining and data mining, including association rule mining have not been used for organizational discovery and evaluation in crisis management, and that no work has investigated organizational discovery from formal experience feedback elaborated by the authorities. However, we believe that combining these fields could be beneficial for experience feedback document analysis in crisis management notably according to an organizational perspective.

From an agent-based organizational point of view, we distinguish works addressing organizational structure discovery from those addressing organizational structure evaluation. For instance, authors in [27] used a genetic algorithm-based approach as a solution for organizational structure discovery. They constructed a knowledge-oriented teamwork and defined the organization, first in terms of *tasks*, *members*, and *structure* that represents the relationships, responsibilities, authorities and communications between the members. Then, they complete the organization in terms of matching and support, which respectively refer to the appropriate assignment of organization members to tasks and to the appropriate deployment of members to organization positions. This is supposed to improve the mutual coordination and knowledge sharing between them. Organizational structure evaluation was mainly addressed by Grossi et al. in [3]. Authors focused on the evaluation of an organizational structure modeled as a MAS, introducing a rigorous framework that allows analysis, evaluation and comparison of organizational structures. First, it uses quantitative concepts and metrics on top of graphs for representing the relations of power, coordination and control between agents of a MAS. Then, it uses these metrics to perform a numerical analysis of some characteristic properties such as robustness, flexibility and efficiency of the organizational structure of this MAS. In this study, we use and adapt this framework and propose algorithms to transform text describing experience feedback into a graph compliant with this framework.

Another interesting work is the one presented in [2]. This work both addresses organizational structure discovery and evaluation in the context of crisis management. It introduces two complementary evaluations of disaster management plans, necessarily the process and the organization set up in such situations. It examines the tsunami resolution plan of Ho Chi Minh City, Vietnam as a case study, and defines manually the possible response scenarios. From these scenarios, it produces the corresponding log

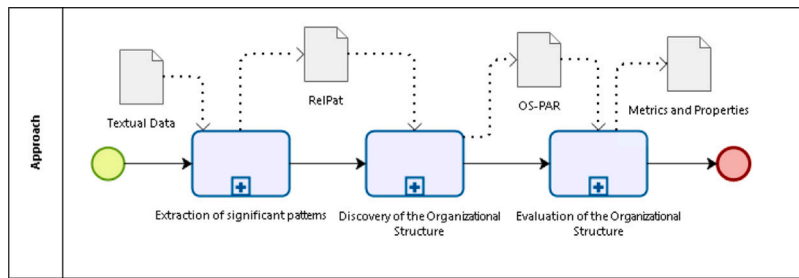


Fig. 1. Approach principle as a BPMN process.

file of activities and discovers the underlying organizational structure from this log file using Alpha, a classical process mining algorithm [28]. Finally, based on the framework given by Grossi et al. [3], it studies and analyzes first the agent organizational structure involved in the resolution plan using the power, coordination and control relations between each pair of roles. Then, it evaluates its robustness, flexibility and efficiency.

The conclusion that can be drawn from this brief overview of the literature is as follows. First, the problem of analyzing experience feedback has received little attention in the field of crisis management. In this work, we address this issue by analyzing textual experience feedback to discover the organization put in place on the ground to deal with a crisis, and by evaluating this organization. It will be thus possible to improve the response plans used for future crises so that they deploy better quality organizational structure with regard to robustness, flexibility and efficiency. Second, text mining was used in several research areas, such as medical area [29,30], and was combined with data mining techniques (including association rule mining) to extract useful information. Nevertheless, both fields were rarely used together to analyze organizational structure in crisis management. The novelty of our work is then to consider these limitations and to close this gap by extracting association rules to find information from feedback recorded in textual documents, to build/discover and visualize the corresponding agent-based organizational structure, and to assess the performance of the discovered structure. Third, the closest work to ours is [2]. However, the approach followed in [2], assumes that the analyzed documents are structured response plans manually treated and transformed into a log-file before being treated to discover the organizational structure. In our work, not only we analyze textual experience feedback automatically but we also discover the organization structure automatically. We start from scratch, by firstly analyzing the text, then extracting the organizational structure using association rule mining and the OSminer algorithm, which is an important and new contribution of our work, and finally by evaluating the organizational structure. Moreover, we discover and evaluate real (i.e., concrete) organizational structures deployed in the field, while [2] evaluates abstract (theoretical) organizational structures discovered from structured response plans. To sum up, there have been very few works producing tools to analyze automatically crisis textual feedback in order to improve future crisis management from an organizational point of view. There is also a lack of joint work between text mining and association rule mining in crisis management. In this paper, we argue that the combination of text mining and association rule mining is a promising approach to analyze textual crisis feedback in order to discover and evaluate organizational structures deployed in such contexts.

3. A new approach for organizational structure discovery and evaluation

This section introduces first, as a process, the overall approach we recommend to address the issue of discovering and evaluating a multi-agent organizational structure in the crisis domain and then details the different tasks of this process.

3.1. Overview of our approach

The approach we have defined to address the issue is given in Fig. 1 as a BPMN process [31]. The dataset used as input is a textual document in which a response to a real crisis is described. The proposed BPMN process consists in three main and different tasks, starting from the "Extraction of significant patterns", passing through the "Discovery of the organizational structure", until the "Evaluation of the organizational structure". These tasks will be described respectively in Sections 3.2, 3.3 and 3.4.. Specific BPMN data objects have been introduced to model information consumed and produced by these tasks (e.g., Textual Data, RelPat, OS-PAR, Metrics and Properties). The process activities and the data objects show the steps followed to move from a discursive and textual description of an organization to its evaluation in the form of metrics (completeness, economy, ...) and properties (robustness, efficiency, flexibility), including an intermediate graph representation (OS-PAR). More precisely, the "Textual Data" corresponds to the textual experience feedback from a past crisis, "RelPat" is a set of relevant patterns discovered from the "Textual Data", OS-PAR is a set of association rules describing organizational patterns represented by a graph. OS-PAR is discovered by our OS-Miner algorithm.

3.2. Step 1: extraction of significant patterns

Step 1.1: Pre-processing of textual data. Cleaning the corpus requires a couple of transformations, namely the removal of punctuation/numbers, supplemental forms (as transition words, or connectors, most common words) and white spaces.

Step 1.2: Extraction of frequent patterns/words. To compute the frequency of the patterns, we firstly construct a paragraph-word matrix (PWM) that lists all occurrences of the words in the corpus, by paragraph. The paragraphs are represented by rows and the words (or patterns or terms) by columns. If a word occurs in a particular paragraph, then the matrix entry corresponding to that row and column is 1, otherwise it is 0. The frequency values of the patterns is derived from 'PWM', by computing the sum of occurrences of each pattern in all row paragraphs of PWM, i.e., it is equal to the sum of all occurrences of the word in the corpus.

Step 1.3: Identification of relevant patterns. The selection of relevant patterns is not an easy step. It requires an in-depth analysis of the semantic of the extracted patterns to only keep the most interesting ones. In collaboration with an expert who was member of the considered crisis cell, we proceed as follows:

- selection of patterns with a frequency higher than a user-defined threshold t_w and generation of a reduced set of frequent patterns, called 'FreqSet'.
- adverbs and adjectives are discarded from 'FreqSet' because they are judged as insignificant for this study.
- selection of a unique pattern to represent a set of words (as verbs and nouns) judged as semantically equivalent and combination of their binary data.
- identification of patterns that differ semantically but may be combined to represent only one pattern, as the name of the actors, or cities.
- elimination of patterns judged as irrelevant by the expert, as unimportant actors or patterns labeled as semantically unknown.

Accordingly, we generate a new set, 'RelPat', of most important and semantically relevant patterns.

3.3. Step 2 : discovery of the organizational structure

Semantic classification of significant patterns. We classify semantically all the discovered patterns of 'RelPat' according to the three semantic relations [3] of coordination, control, and power. Each generated category of patterns describes a small collection of words that enable understanding the way in which actors interact regarding a specific relation. For example, from a sentence such as "A Firefighter informs the Police", we derive a coordination relation between both actors since the verb "inform" is related to coordination. In the same way, verbs such "activate" and "exchange" refer respectively to "power" and "control".

Extraction of important association rules. We then use association rules as a data mining solution to discover possible interactions between the actors. For that, we rely on one of the classical association rules extraction algorithms [12], and on the ' PWM_k ' binary matrix corresponding to each group of patterns (or each relation k) of the 'RelPat'. Multiple constraints are considered when selecting only most important association rules, as:

- the frequency and confidence values of an association rule must be higher than user-defined minimum thresholds t_f and t_c , respectively, in order to extract exact and rare rules.
- the presence of at least one actor C both in the premise and conclusion of each association rule,
- the presence of at least one word in the premise and/or conclusion of each association rule.

Since we are looking to discover how actors communicate together in the crisis process, we judge that all the rules that do not include at least one actor are discarded. Only association rules with the 3 following forms are retained: " $Actors, Words \rightarrow Actors$ "; " $Actors \rightarrow Actors, Words$ "; and " $Actors, Words \rightarrow Actors, Words$ ".

Each rule must have at least one actor in the premise and conclusion and one word in the premise and/or the conclusion. The derived set of association rules is called AR_k , where k in $\{power, control, coordination\}$.

Generation of the organizational structure: the OS-miner algorithm. To generate the organizational structure, two main steps are considered. First we define an organizational structure based on pattern association rules (OS-PAR) corresponding to the three relations k , and then we visualize the defined organizational structure.

Definition of the OS-PAR. Actually, we define an organizational structure-pattern association rule (OS-PAR) as a set of association rules $OSrules_k(x, y)$ from *Selected Rule_k*, with:

$$p_k(x, y) \rightarrow e_k(x, y) \quad (4)$$

where $p_k(x, y)$ represents an actor (or node) in the organizational structure related to the relation k in which x and y are two designed Roles, and $e_k(x, y)$ is an edge or relation labeled e from x to y in the structural dimension k . We refer to p_k and e_k as the premise and conclusion of $OSrules_k$, respectively.

Let us now detail the algorithm 'OSminer' (see Algorithm 1 below) to construct the organizational structure OS-PAR according the three relations of coordination, control and power. 'OSminer' considers as input the set AR_k of most important association rules and focuses on those with at least two actors in each/one side of the rule, to highlight all possible interactions and combinations that may exist between actors in the same paragraph of the textual document. Indeed, it splits the association rules of AR_k set, which have more than one actor in the premise and/or conclusion, into several new association rules (lines 8–12 in the algorithm). Based on the position (premise or conclusion) and the number of actors in the association rule, 'OSminer' generates a new set of derivative association rules of the form " $actor1 \Rightarrow actor2, words$ ". All the words or patterns (present in the premise and conclusion) that describe the relationship between actors are combined together and added to the conclusion of the new rule. For example, if

we have the following coordination association rule: "*prefecture, cod, information* $\Rightarrow$ *spc, dreal, contact*", where *prefecture*, *cod*, *spc* and *dreal* are the actors of the rule, and the pattern *information*, *contact* represent the words of the rule, we obtain as output 4 new derivative association rules with these forms: (i) "*prefecture* $\Rightarrow$ *spc, contact, information*", (ii) "*prefecture* $\Rightarrow$ *dreal, contact, information*", (iii) "*cod* $\Rightarrow$ *spc, contact, information*", and (iv) "*cod* $\Rightarrow$ *dreal, contact, information*". However, for a relation *k* some of the generated rules may have the same actors in the premise and conclusion but different words describing their interactions. We call this set of words "all-combined-words". As a solution, we combine these rules (line 13 in the algorithm) to obtain one unique rule of the form "*actor1* $\rightarrow$ *actor2, all-combined-words*". The application of 'OSminer' algorithm results in the generation of a new set of association rules, *NewRule_k*, where each rule contains only one actor in the premise and conclusion with all their possible interactions under the form of combined words for a relation *k*.

On the whole, probably not all the derived rules from *NewRule_k* are relevant for the study. Some of them may represent an interaction that have never existed between actors in the crisis domain for a given relation. For example, given the relation of power we have the following association rule "*media, order* $\Rightarrow$ *InterdepartmentalStaff*", where *media* gives order to the *InterdepartmentalStaff*. Such rule will never happen because the *InterdepartmentalStaff* is a principal actor who has the power to give orders, contrary to *media*, which is a secondary actor only disseminating information to the public. Accordingly, with the help of the expert we select only association rules that are semantically relevant. We obtain a reduced set of rules w.r.t. each relation, called *SelectedRule_k*.

Visualization of the OS-PAR. The visualization of the multi-agent organizational structure consists in transforming the association rules extracted from the text into a graphical organizational structure. For this, several open source tools for graph and network analysis may be used, as Gephi [32]. Such tools require to adapt the form of the input dataset that in our case is the set of association rules *SelectedRule_k*, in order to generate as output the appropriate organizational structure (OS-PAR).

Algorithm 1: Algorithm to generate new association rules for constructing multi-agent organization

Input: Set of association rules *AR_k* with at least two actors in each/one side of the rule, set of actors *Actors*, set of patterns *RelPat_k* w.r.t. each structure *k*

Output: New set of association rules *NewRule*

1 OSminer begin

2 *NewRule* = $\emptyset$

3 **for** (each relation *k* in 'Coordination', 'Control', 'Power') **do**

4 1. Construct a binary table *WordR* testing the presence/absence of important words from *RelPat_k* in *AR_k*;

5 2. Construct a second binary table *ActorR* testing the presence/absence of *Actors* in each association rule of *AR_k*;

6 3. *premise_k* $\leftarrow$ actors of the left side of each association rule;

7 *conclusion_k* $\leftarrow$ actors of the right side of each association rule;

8 4. Generate new set of association rules *NewRule* with the form *actor1* $\Rightarrow$ *actor2, words*;

9 **for** (*i*; 1 : *nrow*(*AR_k*); *i*++) **do**

10 **for** (*j*; 1 : *length*(*ActorR*); *j*++) **do**

11 **for** (*l*; 1 : *length*(*ActorR*); *l*++) **do**

12 **if** (*premise*[*i*, *j*] == *True* && *conclusion*[*i*, *l*] == *True* && *WordR*[*i*] != ' ') **then**

13 *NewRule_k* $\leftarrow$ *concat*(*premise*[*k*] $\rightarrow$ *conclusion*[*l*], *WordR*[*i*, *l*]);

14 5. Eliminate redundant rules from *NewRule_k*;

15 6. *NewRule* = *NewRule* $\cup$ *NewRule_k*;

16 **return** *NewRule*

3.4. Step 3: evaluation and interpretation of the organizational structure

To evaluate the performance and the relevance of the generated organizational structure, we use the characteristic properties of robustness, flexibility and efficiency [3]. For each property, we compute the scores of a selection of structural evaluation measures, introduced in Table S.1. Then, we compare and evaluate these computed scores with those that maximize each of the three properties, as given in Table 3 in bold. That enables verifying if the OS is robust, flexible and/or efficient. However, relying only on these structural measures and properties may not be sufficient and additional complementary metrics based on the graph theory could be adapted and used in order to have a more precise evaluation study that considers all the structural levels of the organization. To this end, we present in Table 1 a selection of complementary metrics specific to graph-theoretical aspects of an OS which are highly used for graph evaluation. We also give a new formulation to there metrics according to the multi-agent OS notation since they were never used to evaluate an OS. Note that we have not challenged Grossi's framework philosophy, which evaluates the OS at the structural level (without taking into account the role occurrences), and which is well appropriate to the crisis context. We mainly formulate metrics based on centrality which could be used to estimate the influence of a role in an OS, such as: betweenness centrality, closeness centrality and eigenvector centrality [33,34].

Table 1

Re-formulation of a list of metrics -based on the graph theory- to evaluate the OS at Role level. Where $|Roles_k|$ is the number of roles in the organization, A_{xy} is the value of the adjacency matrix between the roles x and $y \in Roles_k$, deg_x is the degree(x), m is the number of interactions of the OS, c designs community, δ is the Kronecker delta which is 1 when roles x and y are assigned to the same community and 0 otherwise, $d(x, y)$ is the distance between two roles $x, y \in Roles_k$, $\lambda \neq 0$ is a constant and $N(x)$ is the set of neighbors of x (in matrix form we have $\lambda x = xA$) and $VLINK_{xy}$ is the edge in relation R_k connecting the role x with role y .

Metric	Definition	Formula	Metric	Definition	Formula
Role degree	It gives the number of interactions m for each role $x \in Roles_k$ in the structural dimension k	$deg_k(x) = m_k(x) $	Betweenness Centrality	It measures the centrality in the organization based on shortest paths, passing through the target role x (denoted $\sigma_{v,y}(x)$).	$BC_x = \sum_{i \neq v \neq y \in Roles_k} \frac{\sigma_{v,y}(x)}{\sigma_{v,y}}$
Role Eccentricity	The eccentricity of $x \in Roles_k$ in a connected structural dimension k is the maximum OS distance $d_k(x, y)$ between x and any other role y of $Roles_k$.	$e_k(x) = \max\{d_k(x, y) x \in Roles_k\}$	Closeness Centrality	It indicates who is at the heart of the OS.	$CC_x = \frac{ Roles_k - 1}{\sum_{y \neq x \in Roles_k} d(x, y)}$
Density	It is equal to the number of its interactions divided by the number of roles present in the organization, considering the three relations.	$density_k(OS) = \frac{ deg_k }{ Roles_k (Roles_k - 1)}$	Eigenvector Centrality	It assesses the "importance" and the "influence" of the roles for the connectivity of the OS.	$EC_x = \frac{1}{\lambda} \sum_{y \in N(x)} A_{x,y} E_y$
Modularity	It is a metric of the structure of networks or organizations. It quantifies the strength of division of an OS into communities.	$Modul(OS) = \frac{1}{2m} \sum_{xy} \left[A_{xy} - \frac{deg_x \times deg_y}{2m} \right] \delta(c_x, c_y)$	Clustering Coefficient	It is defined as the average of the clustering coefficients over all roles in the OS.	$C_k(OS) = \frac{ (VLINK_{xy} : x, y \in Roles_k, VLINK_{xy} \in R_k) }{deg_k(x) \times (deg_k(x) - 1)}$
Diameter	It is the maximum eccentricity of an OS.	$diam_k(OS) = \max\{e(x) x \in Roles_k\}$			
Average Shortest Path Length	It is a measure of efficiency representing the average number of stops needed to reach two distant roles in the structural dimension k .	$l_k = \frac{1}{ Roles_k \times (Roles_k - 1) \times \sum_{(x,y) \in Roles_k} d(x,y)}$			

4. Application in crisis management

In this section, we show and discuss the application of the mining approach to textual experience feedback data from a real crisis situation. The main goal is to discover and evaluate the real organization implemented for crisis resolution modeled as a multi-agent organizational structure that describes the main interactions performed between actors involved in the field, and to propose some improvements if needed.

The textual dataset under analysis describes the global experience feedback of "Pyrénées flood" of June 18th and 19th, 2013 in France.² It is composed of four main sections including in total 2981 words after removal of symbols, numbers and supplemental words. We use the Iramuteq³ software to prepare the dataset for analysis [35]. Actually, we are interested in studying the section specific to the management of the crisis, that describes the response provided on the field by the actors involved in resolving the crisis. This section also fits to our research goal by reporting all the immediate actions taken during the flood event, which may help us identifying relevant ground indicators and improving the crisis process based on a real crisis situation. Contrary to the remaining categories that mainly focus on the chronology, consequences and impacts of the flood event.

Experiments were carried out on using Iramuteq tool as well as Rstudio (Version 0.98.1103, R 3.1.1), two freely available tools enabling textual data analysis. Rstudio offers a selection of packages suitable for such data. We also use Gephi for the visualization of the discovered OS, and for the computation of the additional metrics.

4.1. Pattern mining

Preprocessing. We analyze solely the *crisis management* section from the whole text report, and use the Iramuteq software for preprocessing purpose. A list of 1666 patterns or words are generated, from which 268 patterns are eliminated during the cleaning step. A new dataset of 1398 patterns is then used as input for the extraction of frequent patterns.

² The corresponding file can be downloaded from the following link http://www.occitanie.developpement-durable.gouv.fr/IMG/pdf/Rex_global_cle7db263.pdf.

³ IRAMUTEQ (R INTERFACE for multidimensional analysis of texts and questionnaires). It is a python application anchored in the R software, which enables statistical analysis and different processing of text corpus.

Table 2
Results of the association rules extraction process w.r.t. the three relations.

	R_{Pow}	R_{Ctrl}	R_{Coord}	Total
$RelPat_k$	9	8	15	32
AR_k	84	142	361	587
$NewRule_k$	323	365	400	1088
$SelectedRule_k$	26	9	95	130

Frequent patterns. To identify the set of frequent patterns, we derive the PWM of the 1398 patterns using Iramuteq software. We obtain a binary data table of 47 paragraphs $\times$ 1398 patterns, considered as the starting point to study the frequency of the patterns. The most frequent patterns represent those that are present in all the paragraphs, and then have a frequency equal to 47. The results show that only the pattern ‘Contribution’ has a maximal frequency equal to 47; that 459 patterns have a frequency between 3 and 46; and that an important number of rare patterns equal to 938 have a frequency lower than 3. To keep an adequate number of patterns to work with, and to only extract the frequent ones, we chose to select a frequency threshold equal to 3.

Relevant patterns. To extract only relevant patterns, we go through different tasks as described previously. We set a low frequency threshold $t_w = 3$, which discards an important number of 938 patterns. Hence, we are in presence of a new binary data table (PWM) of 47 paragraphs $\times$ 460 words, where $FreqSet = 460$. The set ‘FreqSet’ of 460 words includes in total 62 adjectives, 11 adverbs, and 18 unknown words (as ‘rt’, ‘st’) which are excluded since they are judged as insignificant for this study. The remaining patterns include consequently verbs and nouns, among which raises the issue of “pattern semantic redundancy”.

To overcome this problem, we propose to first combine all the patterns which are semantically equivalent and keep only one unique word, representative of this category. For example, the verb ‘to contact’ and its corresponding noun ‘contact’ may be combined and represented by one unique pattern called ‘contact’. In total, 42 patterns are judged as semantically equivalent from which 20 words are eliminated for combination purposes. Besides this semantic redundancy issue, we are also facing the problem of identification of patterns – essentially nouns – which are semantically different but occur often together to represent an actor or a city in the corpus. As an example, we have these two nouns ‘Meteo’ and ‘France’ that refer to the pattern ‘Meteo France’, which is an actor in the organization (actors are defined in Table S.2). As a solution, we propose to rely on the help of the expert to specify the exact names of the different actors who participate in this crisis based on the remaining nouns. Only most important actors of the organization are selected from which 8 actors are identified from the combination of at least two nouns, which are: Minister of Ecology, Minister of Interior, Interdepartmental staff, Minister of Defense, City Hall, Meteo France, COD (operational center at the department level), COZ (operational center at the regional level). In total 35 actors are retained after the combination step. In addition to the set of actors, 14 nouns may be combined to represent geographic areas as ‘Midi Pyrénées’, but we decide to ignore these nouns since the name of the cities is not essential for the construction of the OS.

Regarding the identification of the interactions between the actors, unfortunately, a small number of words is obtained. For that reason, we decide to select additional patterns (only verbs) from the set of words with a frequency lower than 3 (16 words are selected). As a consequence, we obtain a new reduced set of 39 patterns that include 10 nouns that correspond to actors and 29 verbs that correspond to relations between actors.

4.2. Discovery of the organizational structure

Semantic classification w.r.t. the relations of power, coordination and control. To classify semantically the remaining patterns (actors are excluded) according to the relations of power, coordination and control, we create three dictionaries of words representative of each relation. We then classify the 39 patterns into three categories regarding the three relations as follows: (i) $RelPat_{coord} = 22$ patterns related to the relation of coordination, (ii) $RelPat_{ctrl} = 8$ patterns related to the relation of control; and finally (iii) $RelPat_{pow} = 9$ patterns related to the relation of power.

Extraction of important association rules. On the basis of the binary data tables relative to each category of patterns ($RelPat_{coord}$, $RelPat_{ctrl}$ and $RelPat_{pow}$), each combined with one of the 35 actors, we apply the mining algorithm of the ‘ConExp’ tool [36] to generate association rules. For all experiments, we set the same support and confidence thresholds values, which are respectively equal to $t_f = 0.1$ and $t_c = 1.0$. With a low support threshold, we do not discard association rules that may describe rare interactions between actors; and with a maximum confidence threshold, we only select exact association rules. The results are described in Table 2 for each relation starting from the number of patterns ($RelPat_k$), passing through the number of extracted association rules (AR_k), and the number of derived ones using “OSminer” ($NewRule_k$) until the final number of selected association rules ($SelectedRule_k$).

Visualization of the organizational structure. We use the Gephi tool [32] to visualize the organizational structures. We start visualizing the big OS of 130 rules that considers the three relations. To improve the visualization, we remove for each given rule the complementary words and we keep only the type of interaction (power, control or coordination) as label and the actors (or roles). We also discard redundant rules (i.e. rules having the same actors in the premise and conclusion for the same relation) resulting from the elimination of the complementary words. Accordingly, only one directed relation is retained between two roles. It results a new reduced set of 35 roles and 148 relations in total.

We compute the Average Weighted Degree of the whole graph or organization, which is ≈ 4 , as well as the degree of each actor. We then retain only actors with degree higher than the average weighted degree equal to 5. In total, 13 actors are eliminated as

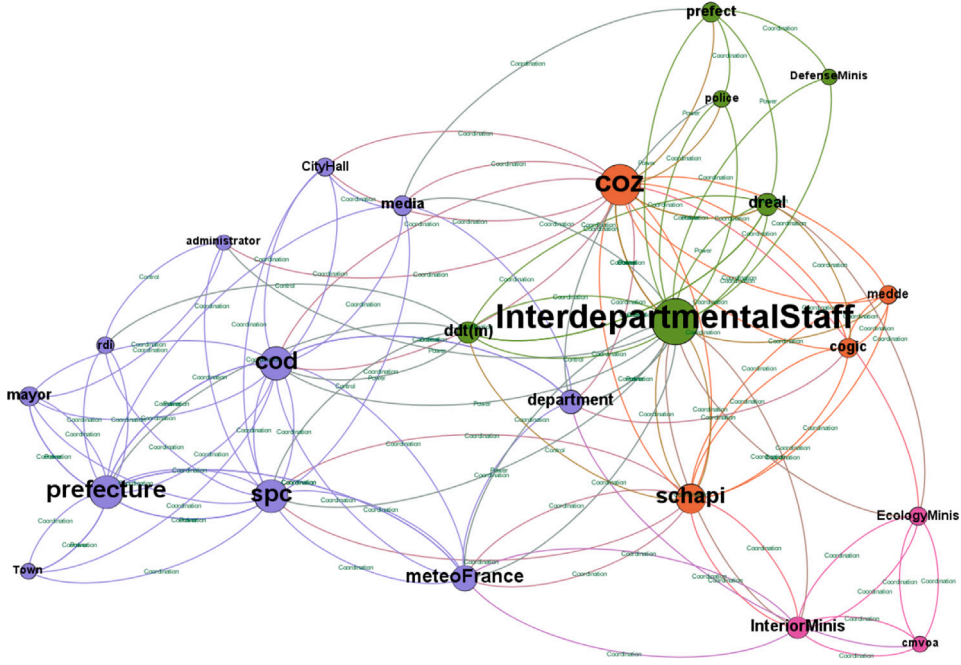


Fig. 2. The organizational structure of 24 actors and their 123 interactions w.r.t. the three relations of coordination, control and power. Each color in the OS defines a community of roles mostly interacting together. The *InterdepartmentalStaff* is the central actor of the organization.

well as their corresponding interactions with other actors in the structure. Fig. 2 presents the new OS of 24 actors (defined in Table S.2) and 123 interactions, labeled with one of the three relations of coordination, control and power.

Fig. 3 presents the OS specific to the relation of power (OS_{pow}). It includes in total 12 roles and 14 interactions after combining together all rules having the same roles in the premise and conclusion but different complementary words (those depicted from different paragraphs). In this figure, we label the interactions with the words related to the power relation (as affect, reinforce, activate, etc.) to highlight how both roles communicate. More precisely, the OS_{pow} shows how powerful actors of the crisis interact with the others. For example, it looks that the *InterdepartmentalStaff* is the main and most powerful actor of the OS, who only gives orders. Some of the other roles both receive and give orders as *prefecture* and *spc*. Others only receive orders as *cod* or *dreal*.

4.3. Evaluation and interpretation of the organizational structure

The discovered OS is a small ($Diameter = 5$ and $Weighted Degree = 5.125$) organization of 24 roles and 123 interactions that only considers the most important and central part of the flood crisis management according to the three relations of coordination, control and power. Based on these relations, we distinguish the global view of the OS from the local ones. The global view considers the three relations at the same time while local views only focus on a single one. In our case, OS_{pow} is a local view that includes 12 roles and 14 interactions (cf. Fig. 3). In the same way, OS_{Coord} includes 24 roles and 100 interactions and OS_{Ctrl} includes 10 roles and 8 interactions.

The global OS reveals weak density value ($Density=0.223$) that could be biased because of the lack of role occurrences in the OS. Following Grossi's framework, we keep the type of the relation existing between two roles and not the number of interactions. More precisely, several interactions may exist between 2 roles with same relation type. For example, 5 association rules are generated between *coz* and *cod* describing directed interactions related to the coordination structure, which are: (r1) *coz* → *cod*, *communication* (r2) *cod* → *coz*, *communication* (r3) *cod* → *coz*, *coordination* (r4) *coz* → *cod*, *contact*, *exchange* and (r5) *cod* → *coz*, *contact*, *exchange*. In our case, we only keep 2 interactions having different directions between both roles labeled *coordination*.

To judge the efficiency, robustness and flexibility of the OS, we rely on the structural metrics introduced by [3]. Table 3 recalls these metrics and presents their computed scores in column Value(OS) and their optimum ones in bold. The comparison of both scores states that the discovered organization has an efficient structure but is not enough flexible nor robust. According to Grossi, it is not possible to maximize all the properties of an OS at the same time [3]. For instance, the efficiency requires a maximal value of $Economy_{Coord}$, contrarily to robustness and flexibility that hurt for minimum value for the same measure. Moreover, flexibility and robustness are compatible: they share three metrics, which must have the same maximal values. In the following, we consider each of these properties.

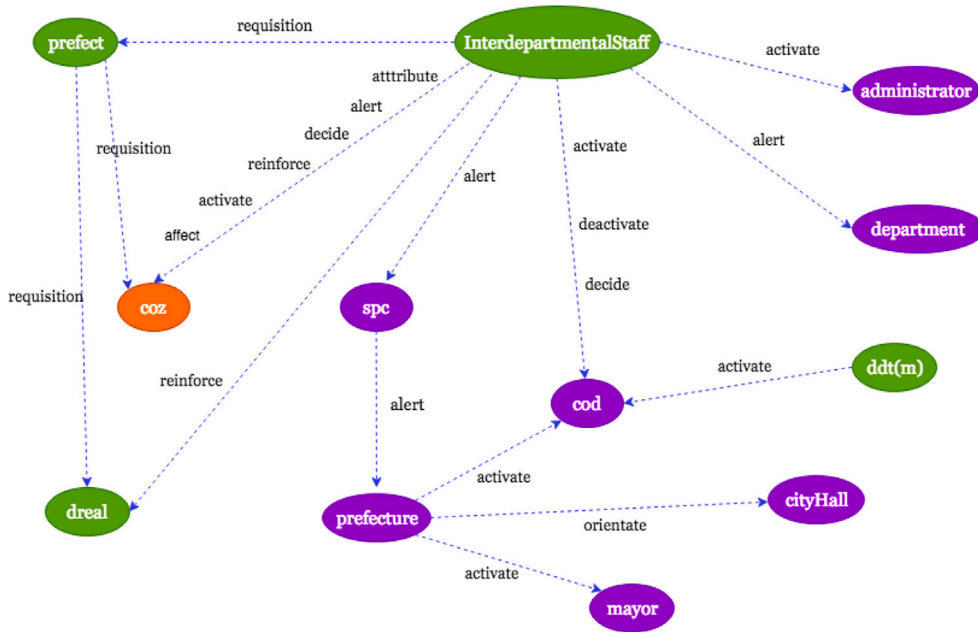


Fig. 3. The organizational structure related to the relation of power with 12 roles and 14 interactions.

The efficiency of the OS. As indicated in Table 3, the efficiency of an OS requires a maximal Economy values according to all relations. The discovered OS approaches the optimal efficiency. It incorporates an efficient power structure with quite optimal values for unilaterality and economy (~ 1). The $Economy_{Control}$ score is equal to the optimal and the $Economy_{coordination}$ is approaching the optimal. Besides that, the low value of the shortest path metric ($Shortest\ Paths(OS) = 2.234$) shows that most of the roles in the organization could communicate easily together. In average, there is only one intermediate agent between two roles, which contributes in the efficiency of the OS.

The robustness of the OS. As indicated in Table 3, the robustness of an OS requires essentially a highly connected coordination structure, a weakly univocal power and control structures, and a weakly unilateral coordination structure. Unfortunately, the discovered OS presents insufficient level of robustness because of its average values for most of the recommended metrics, as for univocity and unilaterality. In addition, not all the roles of the organization are connected together w.r.t the coordination structure ($completeness_{coordination} = 0.18$). The OS is decomposed into four main communities according to the modularity class metric of Table 4, besides its average value for the clustering coefficient metric (equal to 0.40). These communities gather roles mostly connected together than to the rest of the OS.

The first community CM_1 , colored in orange (Fig. 2) has in total 4 roles {schapi, medde, coz and cogic}. Its central and most influential role is coz that has an optimal value for eigen centrality (EC) and the highest value in the community for betweenness centrality (BC). Coz occupies the role of central coordinator at the regional level (with indegree=13.00 and outdegree=8.00). It exchanges information with internal and external roles, i.e. roles inside and outside the community. It also contributes to making internal communication more effective.

The second community CM_2 , colored in green, has 6 roles {prefect, drealm, ddt(m), InterdepartementalStaff, police and DefenseMinis}. The most central role of the community is InterdepartementalStaff since he has the highest closeness centrality (CC). He head the crisis and corresponds to the cell driving and controlling the crisis resolution. InterdepartementalStaff is also a very important role to the flow of information through both the community and the OS due to his highest values in the whole organization for BC and degree. The incoming interactions to the InterdepartementalStaff role are mainly related to the relation of coordination, contrarily to the outgoing interactions, which are more related to the relation of power (see Fig. 3). In this community, the ddt(m) has a frontier role with an average value for CC. Internally, it communicates mainly with InterdepartementalStaff and drealm.

The third community CM_3 , colored in purple, is the biggest community with 11 roles {spc, meteoFrance, cod, prefecture, department, mayor, town, cityHall, rdi, administrator and media}. Its most influential and important role is cod (with the highest BC and EC values in the community). Cod occupies the role of central coordinator at the department level under the responsibility of the prefecture, who has very close values to cod for BC and CC. The spc is another important role in the community as it gives flood forecast to the cod and the prefecture for bidaily decision-making. Meteo-France and media who are at the heart of the community (with highest CC value) have a good position in initiating the spread of information related to forecast updates through the community. Both roles communicate directly with to three roles who control the community (cod, spc and prefecture).

Table 3

Evaluation of the whole OS according to the 3 properties of Efficiency, Robustness and Flexibility. Effective values are given in column “Value (OS)” and the optimal values are given after in bold.

Metric	Value(OS)	Efficiency	Robustness	Flexibility
<i>Economy_{Pow}</i>	0.96	1	–	–
<i>Economy_{Coord}</i>	0.85	1	–	–
<i>Economy_{Ctrl}</i>	1.00	1	–	–
<i>Unilaterality_{Pow}</i>	1.00	1	–	–
<i>Univocity_{Pow}</i>	0.50	1	0	–
<i>Univocity_{Ctrl}</i>	0.40	–	0	–
<i>Unilaterality_{Coord}</i>	0.38	–	0	–
<i>Completeness_{Coord}</i>	0.18	–	1	1
<i>Flatness_{Ctrl}</i>	1.00	–	0	–
<i>Chain_{Pow–Ctrl}</i>	0.30	–	1	1
<i>Chain_{Ctrl–Coord}</i>	0.16	–	1	–
<i>InCover_{Ctrl–Coord}</i>	0.16	–	1	–
<i>OutCover_{Pow–Coord}</i>	0.41	–	1	–
<i>OutCover_{Pow–Ctrl}</i>	0.50	–	1	1
<i>Completeness_{Pow}</i>	0.11	–	–	0

Regardless of his weak values for BC and CC, the *department* has an influential role in the community and the OS because he is directly connected to all important roles of the OS.

The last community *CM₄*, colored in fuchsia, is the smallest one with only three roles {*cmvoa*, *InteriorMinis* and *EcologyMinis*. They have a national role level and are weakly solicited in the context of the considered flood, which has a regional scope.

An optimal robustness would require a complete connectivity between all roles both inside and outside the communities.

The flexibility of the OS. It requires a low score of completeness for the power structure and a high score for the coordination structure to easily adapt. In our case, we have a quite optimal value of completeness for the power relation, equal to 0.11, but a weak value of completeness for the coordination relation, equal to 0.18 instead of 1. Additionally, we have average values of chain and outcover for the control-power relations as given by Table 3. These values confirm that the discovered OS is not enough flexible. This lack of flexibility can also be proven by the values of the eccentricity metric given in Table 4 that highlights the lack of direct connection between two given roles in the OS, and by the ShortestPath metric value. Regarding the connectivity aspect, the shorter the paths are inside the OS, the more flexible it is. The actual OS includes roles that are connected together via intermediary role(s) (ShortestPath=2) and have an eccentricity values between 3 and 5. In addition, many roles are unfortunately not directly connected to the central and decision maker roles of the OS, such as the roles *mayor* and *InterdepartementalStaff* who communicate indirectly via *cod*, making the decision delayed.

The discovered OS has three prominent roles (*coz*, *cod* and *department*) with high EC. They are all directly connected together and to the most popular role of the OS, the *InterdepartementalStaff*. This highlights the connection between communities, where the central role *InterdepartementalStaff* has power relations with the roles *cod* and *coz*, who themselves maintain relations of power and control respectively at the departmental and regional levels.

4.4. Discussion

These results reflect the way crises are managed in France. Indeed, in France, crisis management is under the responsibility of a command and control center, called crisis cell. A crisis cell is headed either by a prefect or by the interior minister, depending on the crisis scale. It is composed of the representatives of different public organizations involved in the crisis resolution. These participating actors are collectively responsible for the crisis resolution: they are responsible for actions undertaken in the field to mitigate risk or deal with damage, and also for the coordination of these actions, which has to be as efficient as possible. In addition, a crisis cell also has to implement resolution plans, which set the framework for the response on the ground. In addition, decision-making is centralized in the crisis cell. This type of management, highly centralized and hierarchical, leads to a lack of flexibility of the OS. Flexibility is a very important property for OS implementing crisis response, due to the evolving nature of crisis and the need to adapt the response to the evolving crisis. Thus, the lack of flexibility must be addressed. In our view, it is mainly due to the lack of direct communication with the crisis cell in which decisions are made, and to the obligation to comply with resolution plans which are regulatory. So, we advocate a more direct communication to the crisis cell to make OS implementing crises response more flexible. We also believe that resolution plans, which must absolutely be used in France, should impose this direct communication. Moreover, we advocate the use of tools making knowledge sharing more effective. Indeed, most of the communication acts in crisis management are related to information sharing. Such tools could then contribute to the reduction of indirect communication acts. This type of management leads also to a lack of robustness of the OS. To address it, we recommend the crisis cells the use of tools making the search for resources easier to perform. The robustness of an OS is linked to its capacity to maintain its functioning, i.e. to maintain the effectiveness of the response on the ground. Finding easily resources to deploy on the ground helps to take better account of this robustness property.

Table 4
Evaluation of the OS roles with regard to 11 measures based on graph theory.

Roles	In degree	Out degree	Degree	Eccentricity	Closeness centrality	Betweenness centrality	Modularity class	Eigen centrality
spc	6.00	11.00	17.00	3.00	0.56	55.67	2.00	0.63
schapi	7.00	8.00	15.00	3.00	0.59	39.82	0.00	0.72
prefect	2.00	4.00	6.00	4.00	0.43	2.23	1.00	0.13
cmvoa	2.00	3.00	5.00	5.00	0.35	0.58	3.00	0.13
medde	3.00	4.00	7.00	4.00	0.47	3.93	0.00	0.36
meteoFrance	7.00	6.00	13.00	3.00	0.56	33.05	2.00	0.76
dreal	3.00	7.00	10.00	4.00	0.49	11.38	1.00	0.23
coz	13.00	8.00	21.00	3.00	0.53	83.93	0.00	1.00
cod	10.00	9.00	19.00	3.00	0.55	79.04	2.00	0.97
cogic	4.00	4.00	8.00	4.00	0.44	1.43	0.00	0.39
ddt(m)	7.00	3.00	10.00	3.00	0.49	22.24	1.00	0.55
prefecture	6.00	12.00	18.00	3.00	0.52	74.96	2.00	0.57
InterdepartmentalStaff	10.00	16.00	26.00	3.00	0.68	151.26	1.00	0.66
police	3.00	2.00	5.00	4.00	0.45	23.26	1.00	0.30
department	9.00	1.00	10.00	5.00	0.32	9.69	2.00	0.95
mayor	5.00	2.00	7.00	4.00	0.40	22.00	2.00	0.52
InteriorMinis	5.00	5.00	10.00	4.00	0.48	30.45	3.00	0.44
DefenseMinis	2.00	1.00	3.00	4.00	0.41	1.07	1.00	0.14
Town	4.00	1.00	5.00	4.00	0.35	0.00	2.00	0.45
EcologyMinis	4.00	3.00	7.00	4.00	0.44	17.38	3.00	0.31
CityHall	4.00	2.00	6.00	4.00	0.40	0.50	2.00	0.45
rdi	2.00	3.00	5.00	4.00	0.40	1.25	2.00	0.20
administrator	4.00	1.00	5.00	5.00	0.29	4.52	2.00	0.42
media	1.00	7.00	8.00	3.00	0.58	11.34	2.00	0.09

5. Conclusion

In this paper, we introduced a new knowledge discovery approach to discover, analyze and evaluate an agent-based organizational structure set up in crisis situation. We proceeded as follows. First, we established a bridge between the fields of text and data mining and multi-agent systems by exploring textual experience feedback recorded in documents, and mining relevant textual patterns useful to extract associations between actors regarding the three relations of power, control and coordination. Second, we presented a new interactive and exploratory approach and proposed the "OSminer" algorithm to enable constructing the agent-based OS, based on the mining of association rules. Third, we extended the work of Grossi and incorporated additional evaluation measures based on graph theory. Finally, we carried out an evaluation study to assess the performance of the discovered organizational structure based on Grossi's framework and on the aforementioned metrics. We mainly examined the key organizational properties of robustness, efficiency and flexibility. In our case study, the discovered organization has an efficient structure because the roles are well connected, but it is not enough robust nor flexible since we miss complete connectivity between all roles. This reflects the way crises are managed in France where the type of management is highly centralized and hierarchical, and can only lead to a lack of flexibility and robustness for the organization. To this end, we have given two main recommendations: (i) regarding the OS flexibility, we think that the more direct the communication to the crisis cell, the quicker the OS reaction time to crisis events; (ii) regarding the OS robustness, we think that faster the crisis cell finds resources, better the OS response capacity is maintained. This work compensates the lack of contributions in the post-crisis stage of the crisis management life cycle. It provides a tool supporting the exploitation of textual experience feedback, the discovery and the evaluation of an OS that drives the resolution of the considered crisis to provide recommendations to better improve preparations for the next crisis.

In future work, we would like to continue this line of research and improve the proposed approach by creating richer dictionaries of words related mainly to actors and actions they made in the organization w.r.t the three relations of power, coordination and control. Moreover, we aim to make the whole exploratory process as much automatic as possible: first we plan to implement the process itself and second to reduce the communication level with the expert of the domain. This reduction is possible thanks to the selection of the top set of association rules based on ranked scores of additional objective measures of interest [37], and not only using support and confidence [18]. We also think to consider the role occurrences to evaluate the OS, which will impact the proposed OSminer algorithm and all metrics given by Grossi. Finally, our work could also be useful in the crisis response stage of the crisis management life cycle, i.e. when crisis cells must respond to a crisis acting on the ground to deal with the damage caused and the risks identified. Indeed, at this stage, the OS set up to respond to the crisis is subject to change and we could assess the impact of the introduction of new roles in the OS w.r.t. the three properties of efficiency, flexibility, and robustness. This is possible if care is taken to specify manually the graph corresponding to the initial OS: the text mining and the discovery steps of our approach should be skipped, and then we could proceed directly to the organization evaluation.

CRedit authorship contribution statement

Dhouha Grissa: Conceived and designed the study, Performed data analysis, Interpretation, Visualization and led and contributed to drafting the manuscript. **Eric Andonoff:** Problem definition, Methodology specification, and also provided valuable

input in drafting and revising the manuscript. **Chihab Hanachi:** Supervised the research leading to this paper, Participated in problem definition and methodology specification, Guided the choice of the agent perspective, and contributed to drafting and revising the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.datak.2023.102237>.

References

- [1] D. Grossi, F. Dignum, M. Dastani, L. Royakkers, Foundations of organizational structures in multiagent systems, in: 4th International Joint Conference on Autonomous Agents and Multiagent Systems, AAMAS 2005, July 25–29, 2005, Utrecht, The Netherlands, ACM, 2005, pp. 690–697.
- [2] L. Thanh, C. Hanachi, S. Stinckwich, T.V. Ho, Combining process simulation and agent organizational structure evaluation in order to analyze disaster response plans, in: KES-AMSTA, in: Smart Innovation, Systems and Technologies, vol. 38, Springer, 2015, pp. 55–68.
- [3] D. Grossi, F. Dignum, V. Dignum, M. Dastani, L. Royakkers, Structural aspects of the evaluation of agent organizations, in: Coordination, Organizations, Institutions, and Norms in Agent Systems II - AAMAS 2006 and ECAI 2006 International Workshops, COIN 2006 Hakodate, in: Lecture Notes in Computer Science, vol. 4386, Springer, 2006, pp. 3–18.
- [4] J. Tukey, *Exploratory Data Analysis*, Addison-Wesley Publishing Company, 1977.
- [5] M. van Leeuwen, Interactive data exploration using pattern mining, in: A. Holzinger, I. Jurisica (Eds.), *Interactive Knowledge Discovery and Data Mining in Biomedical Informatics: State-of-the-Art and Future Challenges*, Springer Berlin Heidelberg, 2014.
- [6] H. Blockeel, Data mining: From procedural to declarative approaches, *New Gener. Comput.* 33 (2) (2015) 115–135.
- [7] Y. Kodratoff, Knowledge discovery in texts: A definition, and applications, in: *Proceedings of the 11th International Symposium on Foundations of Intelligent Systems, ISMIS '99*, Springer-Verlag, 1999, pp. 16–29.
- [8] A. Amihoud, A. Yonatan, F. Ronen, F. Moshe, Maximal association rules: A tool for mining associations in text, *J. Intell. Inf. Syst.* 25 (3) (2005) 333–345.
- [9] H. Mahgoub, Mining association rules from unstructured documents, in: *Proc. 3rd Int. Conf. on Knowledge Mining, ICKM '2006*, 2006, pp. 167–172.
- [10] X. Chen, Y. Wu, Personalized knowledge discovery: Mining novel association rules from text, in: *SDM, SIAM*, 2006, pp. 589–593.
- [11] H. Mahgoub, D.F. Rösner, N. Ismail, F. Torkey, A text mining technique using association rules extraction, *Int. J. Comput. Inf. Eng.* 2 (6) (2008) 2044–2051.
- [12] R. Agrawal, T. Imielinski, A. Swami, Mining association rules between sets of items in large databases, in: *Proc. ACM SIGMOD Intl. Conf. Management of Data*, 1993, pp. 207–216.
- [13] R. Agrawal, R. Srikant, Fast algorithms for mining association rules in large databases, in: *Proceedings of the 20th International Conference on Very Large Data Bases, VLDB '94*, Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 1994, pp. 487–499.
- [14] G. Piattetsky-Shapiro, Discovery, analysis and presentation of strong rules, in: G. Piattetsky-Shapiro, W.J. Frawley (Eds.), *Knowledge Discovery in Databases*, AAAI Press, 1991, pp. 229–248.
- [15] P. Tan, M. Steinbach, V. Kumar, *Introduction to Data Mining*, Pearson Education, 2006.
- [16] S. Sarkar, S. Vinay, C. Djeddi, J. Maiti, Text mining-based association rule mining for incident analysis: A case study of a steel plant in India, *Pattern Recognit. Artif. Intell.* 1322 (2020) 257–273.
- [17] M. Kadhim, M. Alam, Afshar, H. Kaur, A multi-intelligent agent for knowledge discovery in database (MIAKDD): Cooperative approach with domain expert for rules extraction, in: D.-S. Huang, K.-H. Jo, L. Wang (Eds.), *Intelligent Computing Methodologies*, Springer International Publishing, 2014, pp. 602–614.
- [18] A. Ait-Mlouk, F. Gharnati, T. Agouti, Multi-agent-based modeling for extracting relevant association rules using a multi-criteria analysis approach, *Vietnam J. Comput. Sci.* 3 (4) (2016) 235–245.
- [19] M. Tandan, Y. Acharya, S. Pokharell, M. Timilsina, Discovering symptom patterns of COVID-19 patients using association rule mining, *Comput. Biol. Med.* 131 (2021) 104249, <http://dx.doi.org/10.1016/j.compbiomed.2021.104249>.
- [20] A. Borah, B. Nath, Identifying risk factors for adverse diseases using dynamic rare association rule mining, *Expert Syst. Appl.* 113 (2018) 233–263, <http://dx.doi.org/10.1016/j.eswa.2018.07.010>.
- [21] S. Na, S. Sohn, Forecasting changes in Korea Composite Stock Price Index (KOSPI) using association rules, *Expert Syst. Appl.* 38 (7) (2011) 9046–9049, <http://dx.doi.org/10.1016/j.eswa.2011.01.025>.
- [22] N. Uramoto, H. Matsuzawa, T. Nagano, A. Murakami, H. Takeuchi, K. Takeda, A text-mining system for knowledge discovery from biomedical documents, *IBM Syst. J.* 43 (3) (2004) 516–533.
- [23] A. Don, E. Zheleva, M. Gregory, S. Tarkan, L. Auvi, T. Clement, B. Shneiderman, C. Plaisant, Discovering interesting usage patterns in text collections: Integrating text mining with visualization, in: *Proceedings of the Sixteenth ACM Conference on Conference on Information and Knowledge Management, CIKM '07*, ACM, 2007, pp. 213–222.
- [24] P. Wong, P. Whitney, J. Thomas, Visualizing association rules for text mining, in: *Proceedings of the 1999 IEEE Symposium on Information Visualization, INFOVIS '99*, IEEE Computer Society, 1999, pp. 120–125.
- [25] R. Shabbar, S. Madathil, S. Poranki, M. Khasawneh, Analyzing staff experience feedback using text mining, in: *Proceedings of the 7th Annual World Conference of the Society for Industrial and Systems Engineering*, 2018.
- [26] P. Agarwal, R.A. Aziz, J. Zhuang, Interplay of rumor propagation and clarification on social media during crisis events - A game-theoretic approach, *European J. Oper. Res.* 298 (2) (2022) 714–733, <http://dx.doi.org/10.1016/j.ejor.2021.06.060>.
- [27] H. Huang, C. Chen, Emerging organizational structure for knowledge-oriented teamwork using genetic algorithm, *Expert Syst. Appl.* 36 (10) (2009) 12137–12142.
- [28] W. van der Aalst, *Process Mining: Data Science in Action*, second ed., Springer, Heidelberg, 2016, <http://dx.doi.org/10.1007/978-3-662-49851-4>.
- [29] Y. Liu, S. Navathe, A. Pivoshenko, V. Dasigi, R. Dingleline, B. Ciliax, Text analysis of MEDLINE for discovering functional relationships among genes: Evaluation of keyword extraction weighting schemes, *Int. J. Data Min. Bioinform.* 1 (1) (2006) 88–110.
- [30] D. Westergaard, H. Stærfeldt, C. Tønsberg, L. Jensen, S. Brunak, A comprehensive and quantitative comparison of text-mining in 15 million full-text articles versus their corresponding abstracts, *PLoS Comput. Biol.* 14 (2) (2018) 1–16.
- [31] OMG, *Business Process Model and Notation (BPMN)*, Version 2.0, Object Management Group, 2011, URL <http://www.omg.org/spec/BPMN/2.0>.
- [32] M. Bastian, S. Heymann, M. Jacomy, Gephi: An open source software for exploring and manipulating networks, in: *ICWSM, The AAAI Press*, 2009.
- [33] L.C. Freeman, A set of measures of centrality based on betweenness, *Sociometry* 40 (1) (1977) 35–41, <http://dx.doi.org/10.2307/3033543>.

- [34] H. Franks, N. Griffiths, S.S. Anand, Learning agent influence in MAS with complex social networks, *Auton. Agents Multi-Agent Syst.* 28 (5) (2014) 836–866, <http://dx.doi.org/10.1007/s10458-013-9241-1>.
- [35] P. Ratinaud, IRAMUTEQ: Interface de R pour les analyses multidimensionnelles de textes et de questionnaires, 2009, URL Available at: <http://www.iramuteq.org>, computer software.
- [36] S. Yevtushenko, System of data analysis “Concept Explorer”, in: *Proceedings of the 7th National Conference on Artificial Intelligence, KII '2000*, 2000, pp. 127–134.
- [37] D. Grissa, Behavioral Study of Interestingness Measures of Knowledge Extraction (Ph.D. thesis), Blaise Pascal University, Clermont-Ferrand, France, 2013, URL <https://tel.archives-ouvertes.fr/tel-01023975>.

Dhouha Grissa is an Assistant Professor in the Department of Veterinary and Animal Science at the University of Copenhagen, and member of the Bioinformatics and Computation Biology Group at Center for non-coding RNA in Technology and Health. Previously, she held positions as a senior postdoctoral researcher at the Novo Nordisk Foundation-Center for Protein Research (NNF-CPR) in Denmark and as a post-doc in France, with affiliations at the French National Research Institute for Agriculture, food and Environment (INRAE) of Theix and the National Institute for Research in Digital Science and Technology (INRIA) of Nancy. Dhouha received her Ph.D. in Computer Science from Blaise-Pascal University in Clermont-Ferrand, France, in 2013. Her research interests include Data mining, Text mining, Machine learning and Bioinformatics.

Eric Andonoff Associate Professor at the University Toulouse Capitole in France. He is also member of the IRIT (Institut de Recherche en Informatique de Toulouse) Laboratory. He received his Ph.D in Computer Science from the University of Toulouse, in 1992. His research interests include but are not limited to crisis management, business process and multi-agent simulation.

Chihab Hannachi Professor in Computer Science at Toulouse 1 University, and member of the IRIT Laboratory (SMAC Team). He was the head of University Toulouse 1 site of the IRIT laboratory from 2008 to January 2020, and member of the scientific board of Toulouse 1 University from 2005 to 2009. In 2014, he co-founded the international conference ISCRAMMED. His main research interests include the definition of coordination models for Agent Oriented Information, such as organizational structuring, contracting, negotiation and interaction protocols. He has a deep experience with Petri Nets and his fields of applications include crisis management, inter-organizational workflow, or cooperative information gathering.